

11th grade
Term 2 Catch-Up: C6 Required Sample Size (Solutions)

Contents

Contents

Índice general

Contents

Contents

Evaluated criteria

Problems

1. Weekly Spending Sample Size
2. Utilities Spending Sample Size
3. Designer Earnings Sample Planning
4. Ride-Share Interval Width Planning
5. Confidence Level Margin Tradeoff
6. Delivery Cost Precision Upgrade
7. Subscription Margin Reduction Plan
8. Mobile Banking Precision Stages
9. Nonessential Spending Sample Upgrade
10. Side-Income Precision Scenarios
11. Dual Study Precision Comparison
12. Startup Cost Margin Targets

Full Solutions

I Minimal Solutions

7

Evaluated criteria

- C6: Determine the required sample size to achieve a target margin of error.

Problems

P1 1. Weekly Spending Sample Size

A digital-payment study wants to estimate mean weekly spending by first-job workers. Long-run records show $\sigma = 96$ USD. The team wants 95 % confidence with target margin of error $E = 12$ USD. Determine the minimum required sample size.

C6

$$n = \left(\frac{z^* \sigma}{E} \right)^2 = \left(\frac{1,96 \cdot 96}{12} \right)^2 = 245,8624$$

$n = 246$

P2 2. Utilities Spending Sample Size

A youth housing survey estimates mean monthly utilities spending. The long-run standard deviation is known as $\sigma = 54$ USD. The planner requires 90 % confidence and margin of error $E = 6$ USD. Determine the minimum required sample size.

C6

$$n = \left(\frac{z^* \sigma}{E} \right)^2 = \left(\frac{1,645 \cdot 54}{6} \right)^2 = 219,1880$$

$n = 220$

P3 3. Designer Earnings Sample Planning

A freelance-income team wants the mean weekly platform earnings for young designers. Long-run variability is $\sigma = 125$ USD. The team wants a 95 % confidence estimate with margin of error $E = 20$ USD. Extra context says there are 9,800 active designers and last week's median earning was 410 USD. Determine the minimum required sample size.

C6

$$n = \left(\frac{z^* \sigma}{E} \right)^2 = \left(\frac{1,96 \cdot 125}{20} \right)^2 = 150,0625$$

$n = 151$

P4 4. Ride-Share Interval Width Planning

A transportation budget survey estimates mean monthly ride-share spending by university graduates. The known long-run spread is $\sigma = 72$ USD. The required confidence interval width is $W = 18$ USD at 95 % confidence. Determine the minimum required sample size.

C6

$$E = \frac{W}{2} = \frac{18}{2} = 9$$
$$n = \left(\frac{z^* \sigma}{E} \right)^2 = \left(\frac{1,96 \cdot 72}{9} \right)^2 = 245,8624$$
$$n = 246$$

P5 5. Confidence Level Margin Tradeoff

An e-commerce savings study tracks mean monthly discount use by young adults. Long-run standard deviation is $\sigma = 84$ USD. The team wants the same target margin $E = 10$ USD under two confidence levels: 90 % and 95 %. Compute both required sample sizes and compare.

C6

$$n_{90} = \left(\frac{1,645 \cdot 84}{10} \right)^2 = 190,9371 \Rightarrow 191$$
$$n_{95} = \left(\frac{1,96 \cdot 84}{10} \right)^2 = 271,0644 \Rightarrow 272$$
$$n_{95} > n_{90}$$

P6 6. Delivery Cost Precision Upgrade

A food-delivery cost survey already has $n_{\text{current}} = 140$ users and currently reports margin of error $E_{\text{current}} = 11$ USD at 95 % confidence. The organization now wants margin of error $E_{\text{new}} = 8$ USD at the same confidence level. Determine the required total sample size and additional observations needed.

C6

$$\sigma = \frac{E_{\text{current}} \sqrt{n_{\text{current}}}}{z^*} = \frac{11 \sqrt{140}}{1,96} \approx 66,4664$$
$$n_{\text{required}} = \left(\frac{z^* \sigma}{E_{\text{new}}} \right)^2 = \left(\frac{1,96 \cdot 66,4664}{8} \right)^2 \approx 265,2344 \Rightarrow 266$$
$$n_{\text{additional}} = 266 - 140 = 126$$

P7 7. Subscription Margin Reduction Plan

A streaming-subscription survey has $n_{\text{current}} = 121$ and current margin of error $E_{\text{current}} = 9$ USD at 95 % confidence. Management requests a 20 % reduction in margin of error. Find the new target margin, required total sample size, and additional observations.

C6

$$\begin{aligned}E_{\text{new}} &= (1 - 0,20)(9) = 7,2 \\ \sigma &= \frac{E_{\text{current}} \sqrt{n_{\text{current}}}}{z^*} = \frac{9\sqrt{121}}{1,96} = \frac{99}{1,96} \approx 50,5102 \\ n_{\text{required}} &= \left(\frac{1,96 \cdot 50,5102}{7,2} \right)^2 = 189,0625 \Rightarrow 190 \\ n_{\text{additional}} &= 190 - 121 = 69\end{aligned}$$

P8 8. Mobile Banking Precision Stages

A mobile-banking usage survey currently has $n_{\text{current}} = 160$ and margin of error $E_{\text{current}} = 14$ USD at 90 % confidence. The policy team wants two staged improvements: first a 10 % reduction, then a 30 % reduction from the current margin. Determine both required total sample sizes and additional observations.

C6

$$\begin{aligned}E_{10\%} &= (1 - 0,10)(14) = 12,6 \\ E_{30\%} &= (1 - 0,30)(14) = 9,8 \\ \sigma &= \frac{E_{\text{current}} \sqrt{n_{\text{current}}}}{z^*} = \frac{14\sqrt{160}}{1,645} \approx 107,5830 \\ n_{10\%} &= \left(\frac{1,645 \cdot 107,5830}{12,6} \right)^2 \approx 197,5309 \Rightarrow 198 \\ n_{30\%} &= \left(\frac{1,645 \cdot 107,5830}{9,8} \right)^2 \approx 326,5306 \Rightarrow 327 \\ n_{\text{add},10\%} &= 198 - 160 = 38 \\ n_{\text{add},30\%} &= 327 - 160 = 167\end{aligned}$$

P9 9. Nonessential Spending Sample Upgrade

A personal-finance app study has current sample size $n_{\text{current}} = 100$ users. Long-run records indicate $\sigma = 75$ USD for monthly nonessential spending. The confidence level stays at 95 %. The team wants to reduce the current margin of error by 20 %. First recover the current margin of error, then find the new required total sample size and additional observations.

C6

$$E_{\text{current}} = z^* \frac{\sigma}{\sqrt{n_{\text{current}}}} = 1,96 \frac{75}{\sqrt{100}} = 14,7$$
$$E_{\text{new}} = (1 - 0,20)E_{\text{current}} = 0,80(14,7) = 11,76$$
$$n_{\text{required}} = \left(\frac{z^* \sigma}{E_{\text{new}}} \right)^2 = \left(\frac{1,96 \cdot 75}{11,76} \right)^2 = 156,25 \Rightarrow 157$$
$$n_{\text{additional}} = 157 - 100 = 57$$

P10 10. Side-Income Precision Scenarios

A job-market survey currently has $n_{\text{current}} = 144$ respondents, and long-run salary-spread data give $\sigma = 120$ USD for weekly side-income. Confidence level is 90%. The analysts are considering two reductions from the current margin of error: 25 % and 40 %. Recover the current margin first, then compute both required sample sizes and compare.

C6

$$E_{\text{current}} = 1,645 \frac{120}{\sqrt{144}} = 16,45$$
$$E_{25\%} = (1 - 0,25)(16,45) = 12,3375$$
$$E_{40\%} = (1 - 0,40)(16,45) = 9,87$$
$$n_{25\%} = \left(\frac{1,645 \cdot 120}{12,3375} \right)^2 = 256$$
$$n_{40\%} = \left(\frac{1,645 \cdot 120}{9,87} \right)^2 = 400$$
$$n_{\text{add},25\%} = 256 - 144 = 112$$
$$n_{\text{add},40\%} = 400 - 144 = 256$$

P11 11. Dual Study Precision Comparison

Two independent youth-economy studies use 95 % confidence and want the same 30 % margin-of-error reduction. Study A: $n_{A,\text{current}} = 81$, $\sigma_A = 90$ USD. Study B: $n_{B,\text{current}} = 196$, $\sigma_B = 140$ USD. For each study, recover the current margin first, then find the required total sample size and additional observations. Compare which study needs fewer additional observations.

C6

$$E_{A,\text{current}} = 1,96 \frac{90}{\sqrt{81}} = 19,6, \quad E_{B,\text{current}} = 1,96 \frac{140}{\sqrt{196}} = 19,6$$

$$E_{A,\text{new}} = 0,70(19,6) = 13,72, \quad E_{B,\text{new}} = 0,70(19,6) = 13,72$$

$$n_{A,\text{required}} = \left(\frac{1,96 \cdot 90}{13,72} \right)^2 \approx 165,3061 \Rightarrow 166$$

$$n_{B,\text{required}} = \left(\frac{1,96 \cdot 140}{13,72} \right)^2 = 400$$

$$n_{A,\text{add}} = 166 - 81 = 85, \quad n_{B,\text{add}} = 400 - 196 = 204$$

Study A needs fewer additional observations.

P12 12. Startup Cost Margin Targets

An entrepreneurship-finance study currently has $n_{\text{current}} = 225$ records, with known long-run standard deviation $\sigma = 180$ USD for monthly startup operating costs. Confidence level is 95 %. The team wants to evaluate three reductions from the current margin of error: 15 %, 30 %, and 50 %. Recover the current margin first, then compute all new target margins, required total sample sizes, and additional observations.

C6

$$E_{\text{current}} = 1,96 \frac{180}{\sqrt{225}} = 23,52$$

$$E_{15\%} = (1 - 0,15)(23,52) = 19,992$$

$$E_{30\%} = (1 - 0,30)(23,52) = 16,464$$

$$E_{50\%} = (1 - 0,50)(23,52) = 11,76$$

$$n_{15\%} = \left(\frac{1,96 \cdot 180}{19,992} \right)^2 \approx 311,4187 \Rightarrow 312$$

$$n_{30\%} = \left(\frac{1,96 \cdot 180}{16,464} \right)^2 \approx 459,1840 \Rightarrow 460$$

$$n_{50\%} = \left(\frac{1,96 \cdot 180}{11,76} \right)^2 = 900$$

$$n_{\text{add},15\%} = 312 - 225 = 87$$

$$n_{\text{add},30\%} = 460 - 225 = 235$$

$$n_{\text{add},50\%} = 900 - 225 = 675$$

Full Solutions

P1 Problem 1: $n = 246$.

- P2** Problem 2: $n = 220$.
- P3** Problem 3: $n = 151$.
- P4** Problem 4: $n = 246$.
- P5** Problem 5: $n_{90} = 191$, $n_{95} = 272$; 95% needs more observations.
- P6** Problem 6: $n_{\text{required}} = 266$, $n_{\text{additional}} = 126$.
- P7** Problem 7: $E_{\text{new}} = 7,2$ USD, $n_{\text{required}} = 190$, $n_{\text{additional}} = 69$.
- P8** Problem 8: $n_{10\%} = 198$, $n_{\text{add},10\%} = 38$; $n_{30\%} = 327$, $n_{\text{add},30\%} = 167$.
- P9** Problem 9: $E_{\text{current}} = 14,7$ USD, $n_{\text{required}} = 157$, $n_{\text{additional}} = 57$.
- P10** Problem 10: $n_{25\%} = 256$, $n_{\text{add},25\%} = 112$; $n_{40\%} = 400$, $n_{\text{add},40\%} = 256$.
- P11** Problem 11: $n_{A,\text{required}} = 166$, $n_{A,\text{add}} = 85$; $n_{B,\text{required}} = 400$, $n_{B,\text{add}} = 204$; Study A needs fewer additional observations.
- P12** Problem 12: $E_{15\%} = 19,992$ USD, $E_{30\%} = 16,464$ USD, $E_{50\%} = 11,76$ USD; $n_{15\%} = 312$, $n_{\text{add},15\%} = 87$; $n_{30\%} = 460$, $n_{\text{add},30\%} = 235$; $n_{50\%} = 900$, $n_{\text{add},50\%} = 675$.

Parte I

Minimal Solutions

Evaluated criteria

See full solution above.

Problems

See full solution above.

Weekly Spending Sample Size

$$n = 246$$

Utilities Spending Sample Size

$$n = 220$$

Designer Earnings Sample Planning

$$n = 151$$

Ride-Share Interval Width Planning

$$E = \frac{W}{2} = \frac{18}{2} = 9$$

Confidence Level Margin Tradeoff

$$n_{95} > n_{90}$$

Delivery Cost Precision Upgrade

$$n_{\text{additional}} = 266 - 140 = 126$$

Subscription Margin Reduction Plan

$$n_{\text{additional}} = 190 - 121 = 69$$

Mobile Banking Precision Stages

$$n_{\text{add},30\%} = 327 - 160 = 167$$

Nonessential Spending Sample Upgrade

$$n_{\text{additional}} = 157 - 100 = 57$$

Side-Income Precision Scenarios

$$n_{\text{add},40\%} = 400 - 144 = 256$$

Dual Study Precision Comparison

Study A needs fewer additional observations.

Startup Cost Margin Targets

$$n_{\text{add},50\%} = 900 - 225 = 675$$